# HONR 299: Life Alongside AI Course Syllabus

## Course Information

- **Course Number and Title:** HONR 299: Life Alongside AI
- **CRN:** wl.202620.HONR.29900.057
- **Meeting Times:** Monday March 16, 2026 – Thursday, March 19, 2026; 8:30am-5:30pm
- **Instructional Modality:** Face-to-Face
- **Course credit hours:** 2
- **Prerequisites**: none

## Instructor Contact Information

- **Name of the instructor:** Mark Aronson
- **Office Location:** NA
- **Office Phone Number:** (227) 225-2787
- **Purdue Email Address:** maronson@purdue.edu (preferred)
- **Student consultation hours, times, and location:** during the course week, feel free to ask to meet up before or after class hours. Beyond the course week, you can reach me over email. *(Guidelines for face-to-face and/or remote office hours, and/or how to request help outside office hours)*.

## Teaching Philosophy

I think that one of the most important things we do as humans is contemplate difficult questions and that there are many ways one could do this. One could spend a lot of time studying a topic (otherwise known as a Bachelor's degree), one could create new knowledge in a topic (otherwise known as a PhD), one could explain the topic to others (becoming a communicator or lecturer), or one could engage in dynamic conversation with others, which is exactly the goal of this course. I believe that understanding our relationship with AI is going to be one of the most important questions we will grapple with in the coming decades, both individually and as a collective society, and I figured there was no better way to start doing this than in an intensive course format. There will be content I want you to learn, discussions I want you to participate in, and projects I want you to complete, but my real goal for this course is to have us deeply engage in our central question: how ought we to live alongside AI?

## Course Description

The increasing capabilities of artificial intelligence systems have caused humans to question the uniqueness of our contributions to knowledge work, creativity, and even relationships. To build a better understanding of our place in this emerging dyad, this course seeks to conduct a multiscale exploration of both human and artificial intelligence. We'll begin with the biological side, learning how cells use

signaling pathways to process information and how these cellular levels processes build up to the functions of the human brain. In parallel, we'll explore how deep neural networks, which power today's most capable AI models, encode and transform information. We will use this multiscale lens to examine the relationship between biological and artificial information processing, and ask what it means to be a biologically intelligent being in a world of increasingly powerful artificial intelligence?

## How to Succeed in this Course

My biggest expectation is that you come to this course with curiosity and energy. We are here to probe nothing less than the fundamental questions of what it means to be human and how we ought to exist in a world with increasingly powerful AI. I have designed the course to cover a wide range of disciplines and activity types. I know not everything will resonate with everyone, but my hope is you walk in with an open mind to learn as much as you can in the time we have together.

This course is designed to be accessible to students from majors and disciplines, so I have no expectations around you knowing how to code. The knowledge and skills we will touch on along the way - biology, computer science, math, philosophy – are intended to be in service of interrogating the dyad between biological and artificial intelligence. I expect that you will become familiar with these topics, not experts.

## **Learning Resources, Technology & Texts** (student- and public-facing)

Additional Readings

- The list of additional readings will be on Brightspace; however, as we progress through the course, other resources may be added.

Software/web resources

- Word Processor (i.e. MS Word), remember that MS Office is free for all students.
- Python IDE (VS Code is free to use)
- The course GitHub Repository

Brightspace learning management system (LMS)

- Access the course via Purdue's Brightspace learning management system. Begin with the Start Here tab, which offers further insight into the course and how you can be successful in it. It is strongly suggested that you explore and become familiar not only with the site navigation, but also with the content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

## Course Learning Outcomes

Students will:

1. Explain the fundamentals of biological information processing using multiscale descriptions from molecular interactions to cellular signaling pathways to neural networks in the brain
2. Describe the basics of deep neural networks and the principles these networks use to encode information and make decisions

3. Identify unique and shared properties of information processing by comparing biological and artificial approaches
4. Apply a multiscale lens of information processing to evaluate the implications of artificial intelligence for human lives and livelihood
5. Engage in interdisciplinary dialogue bridging biology, computer science, and the humanities

## Assignments

| Day | Category | Task | Points |
|---|---|---|---|
| Day 1 | Reflection | Intro Activity | 5 |
| Day 1 | Participation | Scaling and Emergence | 5 |
| Day 1 | Project | Biological Intelligence | 10 |
| Day 2 | Participation | Data Reduction Activity | 5 |
| Day 2 | Project | Building a neural network | 10 |
| Day 2 | Reflection | Comparing biological and artificial information processing | 10 |
| Day 3 | Reflection | Machine Relations | 5 |
| Day 3 | Participation | Debate: Transhumanist vs Critical Posthumanism | 10 |
| Day 3 | Project | Fictional Story Mapping | 10 |
| Day 4 | Participation | Is AI Making Us Lazy | 5 |
| Day 4 | Project | Final Project | 15 |
| Day 4 | Reflection | Final Reflection | 10 |
| | | Total | 100 |

## **Grading Scale** (student- and public-facing)

In this class, grades reflect the sum of your achievement of learning outcomes throughout the semester. You will accumulate points as described in the assignments portion above, with each assignment graded according to a rubric. At the end of the semester, final grades will be calculated by adding the total points earned and translating those numbers (out of 100) into the following letters (there will be no partial points or rounding). *NOTE: If you use a percentage-based system, be clear about rounding and decimals*.

A+: 97 - 100

A: 93-96

A-: 90-92

B+: 87-89

B: 83-86

B-: 80-82

C+: 77-79

C: 73-76

C-: 70-72

D+: 67-69

D: 63-66

D-: 60-62

F: 59 or below

## Attendance Policy

This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance will be taken at the beginning of each class and lateness will be noted. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform me of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to is not possible, contact me as soon as possible by Purdue email or phone. For absences that do not fall under excused absence regulations (see below), this course follows the following procedures:

1. Do not come to class if you are feeling ill but DO email me at maronson@purdue.edu, with the subject line: HONR 299 absence. I do not need details about your symptoms. Just let me know you are feeling ill and cannot come to class. If it is an emergency, please follow the University regulations on medical care (see below).
2. Unless it falls under the University excused absence regulations (see below), any work due should be submitted on time via our course Brightspace.
3. If that day's class involves assessed work such as a test or presentation, you and I will plan if and how you can make up the work, following the assignment guidelines. This plan must be done before the next class period, so again, email me immediately when you know that you will miss the class.
4. The most important consideration in any absence is how it will affect your achievement of the assignment objectives and the course learning outcomes.

For cases that fall under **excused absence regulations**, you or your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or certain types of medical care. The processes are detailed, so plan ahead.

## Course Schedule

| **Spring 2026** | **Brightspace Sections** |
|---|---|
| Day 1 (Monday, March 16) | Introduction and Biology |

| | |
|---|---|
| Day 2 (Tuesday, March 17) | AI |
| Day 3 (Wednesday, March 18) | Human-machine Relations |
| Day 4 (Thursday, March 19) | Final Project and Conclusion |

* Schedule and assignments subject to change. Any changes will be posted in the learning management system.

## Day 1

- Introduction
  - Class Introductions and Reflecting on Humanity Activity
  - Syllabus Review
  - The Bitter Lesson Activity
  - Emergence Activity
- Biological Information Processing
  - Biological Information Processing Lecture
  - Introduction to Python notebook
  - Simulating a neuron
  - Biological Information Processing Project

## Day 2

- AI
  - Data Reduction Activity
  - Artificial Neural Networks and Multilayer Perceptrons
  - Biological and AI Comparison
  - From Multilayer Perceptrons to AI Systems
  - The Algorithmic Doppelganger

## Day 3

- Human-Machine Relations
  - Personal Machine Relation Reflection
  - History of Human-Machine Relations
  - Story Mapping Project
  - Transhumanist vs Critical Posthumanism Debate

## Day 4

- Final Project
  - Is AI Making Us Lazy Discussion
  - Final Project
- Conclusion
  - Final Reflection

## Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on the Academic Resources table on your Brightspace homepage.

## Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in the Academic Resources table on your Brightspace homepage.

## Accessibility

Every member of our course should be able to access, use, and learn from the materials we share. This includes all course-related digital content that you and I share in the course. This approach helps promote equal access for everyone at Purdue and is mandated federally by Title II of the Americans with Disabilities Act (ADA). We will work together to provide this access within our Brightspace course.

- My part, as instructor, is to make sure all course materials shared to Brightspace, such as documents, slides, videos and audio, and images, meet accessibility guidelines and to assist you in making sure anything you share is accessible.
- Your role, as a student, is to make sure anything you post for other students to engage with is also accessible, such as peer grading, peer feedback, and discussion board posts. This expectation is built into all course assignments that require you to post to Brightspace.
- A good starting place for you is to bookmark and review the Innovative Learning Accessibility Checklist for guidance on creating accessible materials.
- When selecting materials to share on our Brightspace from Purdue Libraries catalog or databases, best practices include choosing items that:
  - Can be downloaded in full
  - Are available in EPUB or HTML formats
  - Include alternative text for written materials or captions for audio/visual content

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: How To Use Your Course Accommodation Letter. Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodation.

***Accommodations for disabled students***

- *The Disability Resource Center (DRC) is a key partner in providing services, resources, and programs that enable disabled students at Purdue to fully participate in curricular and co-curricular offerings. Check out the DRC Instructor Resources website for accommodations it provides (including when/how to request testing accommodations), and tips for creating accessible courses and materials.*

  *Please read information on Student Course Accommodation Letters carefully. Note that you may receive a Course Accommodation Letter at any point in the semester through your Purdue email. Each letter will identify a DRC Access Consultant assigned to the student and is your contact if you have questions about the implementation of an accommodation specific to that student. Therefore,* ***do not require students*** *to send their letters to you by a certain date or to speak with you in order to receive an accommodation.*

  ***Testing accommodations*** *are granted by the DRC to students whose condition may impact their equal access to learning assessments. To ensure this accessibility it is the responsibility of students to request accommodations, the responsibility of instructors to fulfill accommodations and Purdue Testing Service's responsibility to be a resource for both parties. See Instructor Resources for Accommodated Testing - Purdue Testing Services for details. Courses at Purdue in Indianapolis can seek testing support from the onsite DRC staff.*

  *The DRC website provides a sample syllabus statement as well as other Instructor Resources such as the Usable Materials Center. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.*

- *The Faculty Accessibility Consultant in the Center for Instructional Excellence, Dr. Leslie Miller, directly supports faculty and instructors as they implement students' required accommodations as determined by the DRC.*

## Mental Health/Wellness Statement

**If you need support and information about options and resources**, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

## Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

See Purdue's Information on Emergency Preparation and Planning. This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the *Emergency Preparedness for Classrooms document*.

The first day of class, I will review the **Emergency Preparedness plan for our specific classroom**. Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm.

- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat such as a shooting.

## Student-Related policies

The Vice President for Ethics & Compliance website includes a list of Student Policies. Among those is the Violent Behavior policy, which explains that Purdue is committed to providing a safe and secure campus environment for members of the university community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent behavior impedes such goals. Therefore, violent behavior is prohibited in or on any University facility or while participating in any university activity.

## Basic Needs Program

If you are facing challenges securing basic needs such as food, housing, transportation, health services, or access to technology or childcare resources and believe this may affect your performance in the course, please contact the Office of the Dean of Students (ODOS) to help coordinate with community resources. These services vary by location. In **West Lafayette**, see the Basic Needs Program website, or email basicneeds@purdue.edu. To connect with a Student Support Generalist on the **Indianapolis** campus, contact them by phone at 765-495-7797 or email studentlifeindy@purdue.edu.

## Course Evaluation

Toward the end of this semester, you will be provided with an opportunity to give feedback on this course and your instructor. Purdue uses an online course evaluation system, and I will not have access to this anonymous feedback until after final grades are submitted. You will receive an official email from evaluation administrators with a link to the online evaluation site and will receive a prompt to complete the survey when you login to Brightspace. The subject line will be: *Please take 2-5 minutes to complete the survey*. Check your "Junk E-mail" folder occasionally to be sure the evaluation emails were not accidentally routed there. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.